

compared to the national average of high teens, with several circles reporting losses of 30-40%.

The Adani Energy Solutions Advantage

The smart metering initiative by AESL is built around the government's 10-year DBFOOT (Develop, Build, Fund, Own, Operate, and Transfer) contract model. Adani's current order book stands at 2.28 crore smart meters, with a contract value of over Rs 27,000 crore, representing a market share of 20%.

AESL is working with discoms in Assam, Maharashtra, Andhra Pradesh, Uttarakhand, and Bihar to deploy smart meters, as well as within its own distribution area of Mumbai suburbs, which it serves through its subsidiary Adani Electricity Mumbai Ltd. (AEML). The company has taken an integrated approach towards smart meter deployment to have larger control over the whole value chain through the project lifecycle.

The key elements of a smart meter rollout include the physical meter, a secure communication link between the meter and the discom, trained manpower for installation and troubleshooting, HES & MDM (Head End System/ Meter Data Management) for

controlling meters, a secure cloud network where this data is stored and analyzed by the discoms, and a consumer app enabling them to pay bills, analyze consumption patterns, and receive communication from the discoms.

Recognizing the risks associated with technology and vendor obsolescence over the project lifecycle, AESL has strategically chosen to in-source as much as possible.

AESL has formed a 49:51 joint venture with UAE-based Esyasoft Holdings for smart metering projects in India and beyond. Esyasoft is renowned for its cloud and IoT-based energy management solutions for smart utilities and brings a rich repository of intellectual property in MDM, HES, peak load management, energy audits, metering, billing, mobility solutions, big data analytics, theft analysis, consumer behavior analysis, and GIS.

To further strengthen its control and visibility over the entire smart meter deployment chain, AESL has partnered with sister firms like AdaniConneX for local data hosting and Adani Digital Labs for customizing the consumer app. Through its CSR arm, AESL is also training a workforce of technicians to facilitate a swift rollout and create employability

opportunities within the industry.

It has tied up with Airtel Business – the B2B arm of Bharti Airtel – for reliable and secure connectivity between the smart meters and HES, besides smart meter tracking and monitoring with advanced analytics and diagnostic capabilities and real-time insights and services that empower customers with enhanced control over their energy consumption.

AESL's comprehensive smart metering initiative not only exemplifies technological innovation and strategic partnerships but also underscores a commitment to sustainability and efficiency in energy management. By integrating advanced analytics, secure connectivity, and localized data solutions, AESL is poised to transform the energy landscape, driving significant reductions in transmission and distribution losses. This forward-thinking approach not only empowers utility providers with superior demand forecasting and load management capabilities but also incentivizes consumers to adopt energy-efficient practices. As AESL continues to expand its smart meter deployment across India, it sets a benchmark for the industry, championing a smarter, greener future for all.

